

141. If the solution of the differential equation $(2x + 3y - 2) dx + (4x + 6y - 7) dy = 0$, $y(0) = 3$, is $\alpha x + \beta y + 3 \log_e |2x + 3y - \gamma| = 6$, then $\alpha + 2\beta + 3\gamma$ is equal to _____.

यदि अवकल समीकरण $(2x + 3y - 2) dx + (4x + 6y - 7) dy = 0$, $y(0) = 3$ का हल $\alpha x + \beta y + 3 \log_e |2x + 3y - \gamma| = 6$ है, तो $\alpha + 2\beta + 3\gamma$ बराबर है _____।

Ans. 29

Sol. $(2x + 3y - 2)dx + (4x + 6y - 7)dy = 0$

Let $2x + 3y = t$

$$2 + 3 \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{1}{3} \left[\frac{dt}{dx} - 2 \right] = - \left(\frac{t-2}{2t-7} \right)$$

$$\frac{dt}{dx} - 2 = - \left(\frac{3t-6}{2t-7} \right)$$

$$\frac{dt}{dx} = 2 - \frac{3t-6}{2t-7}$$

$$\frac{dt}{dx} = \frac{t-8}{2t-7}$$

$$\int \left(\frac{2t-7}{t-8} \right) dt = \int dx$$

$$= \int 2dt + \int \frac{a}{t-8} dt = \int dx$$

$$= 2t + 9 \ln |t-8| = x + c$$

$$= 2(2x + 3y) + 9 |2x + 3y - 8| = x + c$$

142. If the solution curve, of the differential equation $\frac{dy}{dx} = \frac{x+y-2}{x-y}$ passing through the point (2, 1) is

$$\tan^{-1} \left(\frac{y-1}{x-1} \right) - \frac{1}{\beta} \log_e \left(\alpha + \left(\frac{y-1}{x-1} \right)^2 \right) = \log_e |x-1|, \text{ then } 5\beta + \alpha \text{ is equal to } \underline{\hspace{2cm}}.$$

यदि अवकलन समीकरण $\frac{dy}{dx} = \frac{x+y-2}{x-y}$ का हल वक्र, जो बिंदु (2, 1) से होकर जाता है,

$$\tan^{-1} \left(\frac{y-1}{x-1} \right) - \frac{1}{\beta} \log_e \left(\alpha + \left(\frac{y-1}{x-1} \right)^2 \right) = \log_e |x-1| \text{ है, तो } 5\beta + \alpha \text{ बराबर है } \underline{\hspace{2cm}}$$

Ans. 11

Sol. $\frac{dy}{dx} = \frac{x+y-2}{x-y}$

$x+y-2=0$ and $x-y=0$ intersect at $(1, 1)$

$x = X + 1, y = Y + 1$

$\Rightarrow \frac{dY}{dX} = \frac{X+Y}{X-Y}$

Put $Y = tX \rightarrow \frac{dY}{dX} = t + X \frac{dt}{dX}$

$\Rightarrow t + X \frac{dt}{dX} = \frac{1+t}{1-t}$

$X \frac{dt}{dX} = \frac{1+t^2}{1-t}$

$\frac{dX}{X} = \frac{1-t}{1+t^2} dt$

$\ell n|X| = \tan^{-1} t - \frac{1}{2} \ln|1+t^2|$

$\ell n|x-1| = \tan^{-1} \left(\frac{y-1}{x-1} \right) - \frac{1}{2} \ell n \left| 1 + \left(\frac{y-1}{x-1} \right)^2 \right| + C$

Passes through $(2, 1) \Rightarrow C = 0$

$\Rightarrow \tan^{-1} \left(\frac{y-1}{x-1} \right) - \frac{1}{2} \ell n \left(1 + \left(\frac{y-1}{x-1} \right)^2 \right) = \ell n|x-1|$

$\Rightarrow 5\beta + \alpha = 11$

143. If the solution curve $y = y(x)$ of the differential equation $(1+y^2)(1+\log_e x) dx + x dy = 0, x > 0$ passes through the

point $(1, 1)$ and $y(e) = \frac{\alpha - \tan\left(\frac{3}{2}\right)}{\beta + \tan\left(\frac{3}{2}\right)}$, then $\alpha + 2\beta$ is _____.

यदि अवकल समीकरण $(1+y^2)(1+\log_e x) dx + x dy = 0, x > 0$ का हल वक्र $y = y(x)$, बिन्दु $(1, 1)$ से होकर जाता है

तथा $y(e) = \frac{\alpha - \tan\left(\frac{3}{2}\right)}{\beta + \tan\left(\frac{3}{2}\right)}$ है, तो $\alpha + 2\beta$ बराबर है _____.

Ans. 3

Sol. $(1 + y^2)(1 + \ln x)dx + xdy = 0$

$$\left(\frac{1 + \ln x}{x}\right)dx = -\frac{dy}{1 + y^2}$$

$$\int \left(\frac{1 + \ln x}{x}\right)dx = \int -\frac{dy}{1 + y^2}$$

$$\tan^{-1} y = -\frac{(1 + \ln x)^2}{2} + C$$

curve passes through (1, 1) $\Rightarrow C = \frac{\pi}{4} + \frac{1}{2}$

$$y = \tan\left(\frac{\pi}{4} + \frac{1}{2} - \frac{(1 + \ln x)^2}{2}\right)$$

$$y(e) = \tan\left(\frac{\pi}{4} + \frac{1}{2} - 2\right)$$

$$y(e) = \tan\left(\frac{\pi}{4} - \frac{3}{2}\right) = \frac{1 - \tan\left(\frac{3}{2}\right)}{1 + \tan\left(\frac{3}{2}\right)}$$

Hence $\alpha = \beta = 1$

$\alpha + 2\beta = 3$

144. Let $f(x) = \sqrt{\lim_{r \rightarrow x} \left\{ \frac{2r^2 \left[(f(r))^2 - f(x)f(r) \right]}{r^2 - x^2} - r^3 e^{\frac{f(r)}{r}} \right\}}$ be differentiable in $(-\infty, 0) \cup (0, \infty)$ and $f(1) = 1$. Then

the value of ea, such that $f(a) = 0$, is equal to _____.

माना $f(x) = \sqrt{\lim_{r \rightarrow x} \left\{ \frac{2r^2 [(f(r))^2 - f(x)f(r)]}{r^2 - x^2} - r^3 e^{\frac{f(r)}{r}} \right\}}$ अंतराल $(-\infty, 0) \cup (0, \infty)$ में अवकलनीय है तथा

$f(1) = 1$ है। यदि $f(a) = 0$ है, तो ea का मान है _____.

Ans. 2

Sol. $f(x) = \sqrt{\lim_{r \rightarrow x} \left\{ \frac{2r^2 [(f(r))^2 - f(x)f(r)]}{r^2 - x^2} - r^3 e^{\frac{f(r)}{r}} \right\}}$

$$\Rightarrow (f(x))^2 = xf(x)f'(x) - x^3 e^{\frac{f(x)}{x}} \text{ by L.H. rule}$$

$$\Rightarrow y^2 = xy \frac{dy}{dx} - x^3 e^{\frac{y}{x}}$$

$$\Rightarrow \frac{y}{x} = \frac{dy}{dx} - \frac{x^2}{y} e^{\frac{y}{x}}$$

Let $y = vx \therefore \frac{dy}{dx} = v + x \frac{dv}{dx}$

$$\Rightarrow v = v + x \frac{dv}{dx} - \frac{x^2}{vx} e^v$$

$$\Rightarrow x \frac{dv}{dx} - \frac{x}{v} e^v = 0$$

$$\Rightarrow ve^{-v} dv - dx = 0$$

$$\Rightarrow (x+c)e^v + v + 1 = 0 \text{ (Integration)}$$

$$\Rightarrow (x+c)e^{y/x} + \frac{y}{x} + 1 = 0$$

$$\therefore y(1) = 1 \therefore c = -1 - \frac{2}{e}$$

$$\therefore \left(x - 1 - \frac{2}{e} \right) e^{y/x} + \frac{y}{x} + 1 = 0$$

$$f(a) = 0 \Rightarrow x = a, y = 0 \therefore \left(a - 1 - \frac{2}{e} \right) + 1 = 0$$

$$ae - e - 2 + e = 0$$

$$ae = 2$$

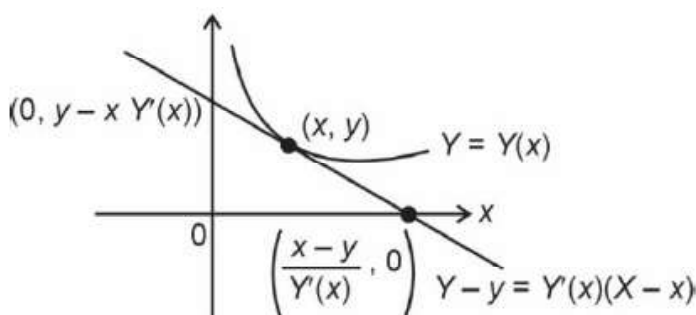
145. Let $Y=Y(X)$ be a curve lying in the first quadrant such that the area enclosed by the line $Y - y = Y'(x)(X - x)$ and the co-ordinate axes, where (x, y) is any point on the curve, is always $\frac{-y^2}{2Y'(x)} + 1$, $Y'(x) \neq 0$. If $Y(1) = 1$, then $12Y(2)$ equals _____.

माना प्रथम चतुर्थांश में एक वक्र $Y=Y(X)$ इस प्रकार है कि रेखा $Y - y = Y'(x)(X - x)$ तथा निर्देशांक अक्षों से घिरे क्षेत्र का क्षेत्रफल

$$\frac{-y^2}{2Y'(x)} + 1, Y'(x) \neq 0 \text{ है, } (x, y) \text{ वक्र पर कोई बिन्दु है। यदि } Y(1) = 1 \text{ है, तो } 12Y(2) \text{ बराबर है } \underline{\hspace{2cm}}$$

Ans. 20

Sol.



$$\Rightarrow \frac{1}{2}(y - xY'(x))\left(x - \frac{y}{Y'(x)}\right) = -\frac{y^2}{2Y'(x)} + 1$$

$$(y - xY'(x))(xY'(x) - y) = -y^2 + 2Y'(x)$$

$$yxY'(x) - y^2 - x^2(Y'(x)) + xyY'(x) = -y^2 + 2Y'(x)$$

$$\left(y - x \frac{dy}{dx}\right)\left(x \frac{dy}{dx} - y\right) = -y^2 + 2 \frac{dy}{dx}$$

$$-y^2 - x^2\left(\frac{dy}{dx}\right)^2 + 2xy\left(\frac{dy}{dx}\right) = -y^2 + 2 \frac{dy}{dx}$$

$$-x^2 \frac{dy}{dx} + 2xy = 2$$

$$\Rightarrow \frac{dy}{dx} + \left(-\frac{2}{x}\right)y = \left(-\frac{2}{x^2}\right)$$

$$\text{I.F.} = \frac{1}{x^2}$$

$$\Rightarrow \text{Solution is } \frac{y}{x^2} = \frac{-2 \cdot x^{-3}}{-3} + c$$

$$\because y(1) = 1$$

$$\Rightarrow c = \frac{1}{3}$$

$$\frac{y}{x^2} = \frac{2}{3x^3} + \frac{1}{3}, \text{ when } x = 2, y = 4 \left(\frac{1}{12} + \frac{1}{3} \right) = \frac{5}{3}$$

$$\Rightarrow 12y(2) = 20$$

146. Let $y = y(x)$ be the solution of the differential equation

$$\sec^2 x dx + (e^{2y} \tan^2 x + \tan x) dy = 0, 0 < x < \frac{\pi}{2}, y\left(\frac{\pi}{4}\right) = 0. \text{ If } y\left(\frac{\pi}{6}\right) = \alpha, \text{ then } e^{8\alpha} \text{ is equal to } \underline{\hspace{2cm}}.$$

माना अवकल समीकरण $\sec^2 x dx + (e^{2y} \tan^2 x + \tan x) dy = 0$, $0 < x < \frac{\pi}{2}$, $y\left(\frac{\pi}{4}\right) = 0$ का हल $y = y(x)$ है। यदि

$$y\left(\frac{\pi}{6}\right) = \alpha \text{ है, तो } e^{8\alpha} \text{ बराबर है } \underline{\hspace{2cm}}$$

Ans. 9

Sol. $\sec^2 x dx + (e^{2y} \tan^2 x + \tan x) dy = 0$

Put $\tan x = t$

$$\sec^2 x dx = dt$$

$$dt + (e^{2y} t^2 + t) dy = 0$$

$$\frac{1}{t^2} \times \frac{dt}{dy} + \frac{1}{t} = -e^{2y}$$

Put $\frac{1}{t} = u$

$$-\frac{1}{t^2} \frac{dt}{dy} = \frac{du}{dy}$$

$$-\frac{du}{dy} + u = -e^{2y}$$

$$\frac{du}{dy} - u = e^{2y}$$

$$\text{I.f.} = e^{-\int dy} = e^{-y}$$

$$u \cdot e^{-y} = \int e^{-y} e^{2y} dy$$

147. If $x = x(t)$ is the solution of the differential equation $(t+1)dx = (2x + (t+1)^4)dt$, $x(0) = 2$, then $x(1)$ equals _____.

यदि अवकल समीकरण $(t+1)dx = (2x + (t+1)^4)dt$, $x(0) = 2$ का हल $x = x(t)$ है, तो $x(1)$ बराबर है _____

Ans. 14

Sol. $(t+1)dx = (2x + (t+1)^4)dt$

$$\frac{dx}{dt} - \frac{2x}{(t+1)} = (t+1)^3$$

$$\text{IF} = e^{-\int \frac{2}{t+1} dt} = \frac{1}{(t+1)^2}$$

$$\frac{x}{(t+1)^2} = \int (t+1) dt$$

$$\frac{x}{(t+1)^2} = \frac{t^2}{2} + t + C$$

$$\text{Now } x(0) = 2$$

$$\Rightarrow C = 2$$

$$\therefore x = \left(\frac{t^2}{2} + t + 2 \right) (t+1)^2$$

$$x(1) = \left(\frac{1}{2} + 1 + 2 \right) (1+1)^2$$

$$= \frac{7}{2} \times 4 = 14$$

148. If $\frac{dx}{dy} = \frac{1+x-y^2}{y}$, $x(1) = 1$, then $5x(2)$ is equal to _____.

यदि $\frac{dx}{dy} = \frac{1+x-y^2}{y}$, $x(1) = 1$ है, तो $5x(2)$ बराबर है _____

Ans. 5

Sol. $\frac{dx}{dy} - \frac{x}{y} = \frac{1-y^2}{y}$

$$\text{Integration factor} = e^{\int -\frac{1}{y} dy} = \frac{1}{y}$$

$$x \cdot \frac{1}{y} = \int \frac{1-y^2}{y^2} dy$$

$$\frac{x}{y} = \frac{-1}{y} - y + c$$

$$x = -1 - y^2 + cy$$

$$x(1) = 1$$

$$1 = -1 - 1 + c \Rightarrow c = 3$$

$$x = -1 - y^2 + 3y$$

$$5x(2) = 5(-1 - 4 + 6)$$

$$= 5$$

Area Under Curve

149. The area of the region $\{(x, y) : |x - y| \leq y \leq 4\sqrt{x}\}$:

क्षेत्र $\{(x, y) : |x - y| \leq y \leq 4\sqrt{x}\}$ का क्षेत्रफल है :

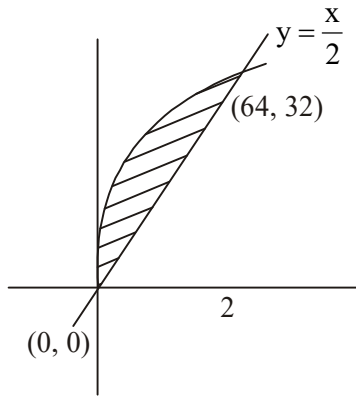
(A) $\frac{1024}{3}$

(B) 512

(C) $\frac{2048}{3}$

(D) $\frac{512}{3}$

Ans. A



Sol.

$$|x - y| \leq y \leq 4\sqrt{x}$$

$$\text{Now, } y = |x - y|$$

$$y^2 = (x - y)^2$$

$$\Rightarrow y = \frac{x}{2} \text{ and } x = 0$$

$$\text{Now, area} = \int_0^{64} \left(4\sqrt{x} - \frac{x}{2} \right) dx$$

$$= \left[\frac{4x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^2}{4} \right]_0^{64} = \frac{8}{3} \cdot 8^3 - \frac{64^2}{4} = 64^2 \left(\frac{1}{12} \right)$$

$$= \frac{1024}{3}$$

150. A line passing through the point A(-2, 0), touches the parabola P: $y^2 = x - 2$ at the point B in the first quadrant. The area, of the region bounded by the line AB, parabola P and the x-axis, is :

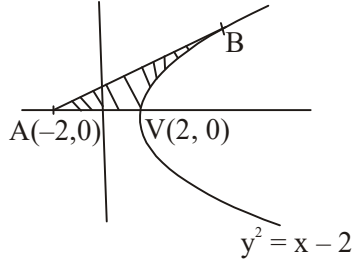
बिन्दु A(-2, 0) से होकर जाने वाली एक रेखा परवलय P: $y^2 = x - 2$ को प्रथम चतुर्थांश में बिन्दु B पर स्पर्श करती है। रेखा AB परवलय P और x-अक्ष से परिबद्ध क्षेत्र का क्षेत्रफल है :

- (A) 2 (B) 3 (C) $\frac{8}{3}$ (D) $\frac{7}{3}$

Ans. C

Sol. Equation of tangent

$$y = m(x - 2) + \frac{1}{m}$$



Passes through $(-2, 0)$

$$0 = -4m + \frac{1}{4m}$$

$$\Rightarrow m = \frac{1}{4}$$

$$y = \frac{1}{4}(x - 2) + 1$$

$$x - 4y + 2 = 0$$

Solving with $y^2 = x - 2$,

$$B(6, 2)$$

$$\text{Area} = \int_0^2 ((y^2 + 2) - (4y - 2)) dy = \frac{8}{3}$$

151. If the area of the region bounded by the curves $y = 4 - \frac{x^2}{4}$ and $y = \frac{x - 4}{2}$ is equal to α , then 6α equals :

यदि वक्रों $y = 4 - \frac{x^2}{4}$ और $y = \frac{x - 4}{2}$ से परिबद्ध क्षेत्र का क्षेत्रफल α है, तो 6α बराबर है :

(A) 250

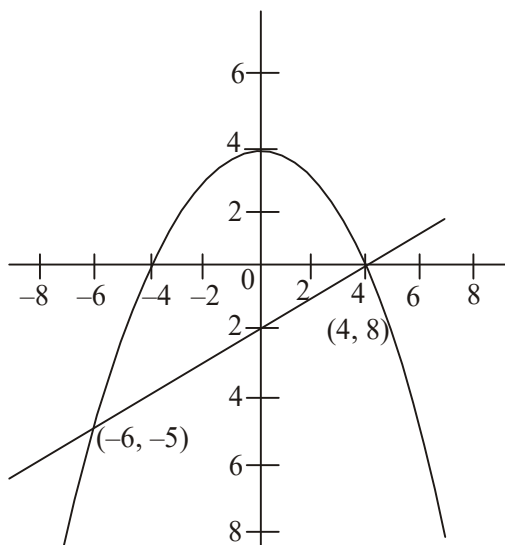
(B) 220

(C) 240

(D) 210

Ans. A

Sol.



$$\text{Area} = \int_{-6}^4 \left\{ \left(4 - \frac{x^2}{4} \right) - \left(\frac{x-4}{2} \right) \right\} dx$$

$$= \int_{-6}^4 \left\{ -\frac{x^2}{4} - \frac{x-6}{2} \right\} dx$$

$$\alpha = -\frac{x^3}{12} - \frac{x^2}{4} + 6x \Big|_{-6}^4 = \frac{125}{3}$$

$$\therefore 6\alpha = 250$$

152. If the area of the region $\{(x, y) : 1 + x^2 \leq y \leq \min \{x + 7, 11 - 3x\}\}$ is A, then 3A is equal to :

यदि क्षेत्र $\{(x, y) : 1 + x^2 \leq y \leq \min \{x + 7, 11 - 3x\}\}$ का क्षेत्रफल A है, तो 3A बराबर है :

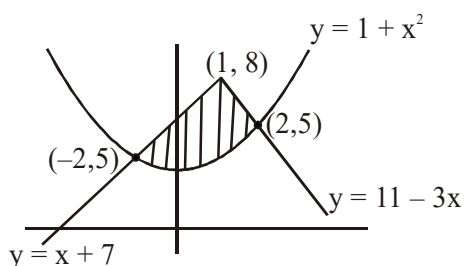
(A) 46

(B) 47

(C) 49

(D) 50

Ans. D



Sol.

$$\begin{aligned}
 A &= \int_{-2}^1 \left((x+7) - (x^2+1) \right) dx + \int_1^2 \left((11-3x) - (x^2+1) \right) dx \\
 &= \left(-\frac{x^3}{3} + \frac{x^2}{2} + 6x \right)_{-2}^1 + \left(-\frac{x^3}{3} - \frac{3x^2}{2} - 10x \right)_1^2 \\
 &= \frac{50}{3}
 \end{aligned}$$

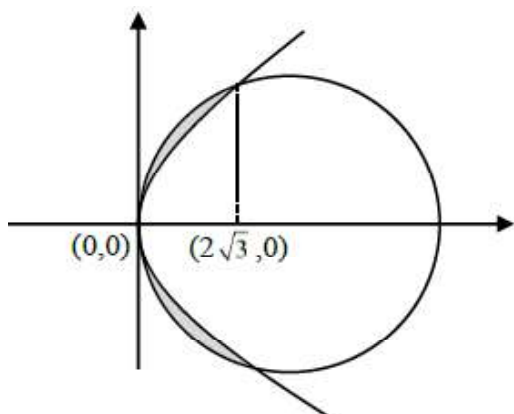
So, $3A = 50$

153. The area of the region, inside the circle $(x - 2\sqrt{3})^2 + y^2 = 12$ and outside the parabola $y^2 = 2\sqrt{3}x$ is :

वृत्त $(x - 2\sqrt{3})^2 + y^2 = 12$ के अन्दर और परवलय $y^2 = 2\sqrt{3}x$ के बाहर के क्षेत्र का क्षेत्रफल है :

- (A) $3\pi + 8$ (B) $6\pi - 16$ (C) $3\pi - 8$ (D) $6\pi - 8$

Ans. B



Sol.

$$y^2 = 2\sqrt{3}x$$

$$(x - 2\sqrt{3})^2 + y^2 = (2\sqrt{3})^2$$

$$A = \frac{\pi r^2}{2} - 2 \int_0^{2\sqrt{3}} \sqrt{2\sqrt{3}x} dx$$

$$\frac{\pi(12)}{2} - 2\sqrt{2\sqrt{3}} \frac{(x^{3/2})_0^{2\sqrt{3}}}{3/2}$$

$$= 6\pi - 16$$

154. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$. If $f'(0) = 4a$ and f satisfies $f''(x) - 3af'(x) - f(x) = 0$, $a > 0$, then the area of the region $R = \{(x, y) \mid 0 \leq y \leq f(ax), 0 \leq x \leq 2\}$ is:

माना कि $f: \mathbb{R} \rightarrow \mathbb{R}$, एक दो बार अवकलनीय फलन इस प्रकार है कि सभी $x, y \in \mathbb{R}$ के लिए, $f(x+y) = f(x)f(y)$ है। यदि $f'(0) = 4a$ है और $f, f''(x) - 3af'(x) - f(x) = 0, a > 0$, को संतुष्ट करता है, तो क्षेत्र $R = \{(x, y) \mid 0 \leq y \leq f(ax), 0 \leq x \leq 2\}$ का क्षेत्रफल है:

- (A) $e^2 + 1$ (B) $e^4 + 1$ (C) $e^2 - 1$ (D) $e^4 - 1$

Ans. C

Sol. $f(x+y) = f(x) \cdot f(y)$

$$\Rightarrow f(x) = e^{\lambda x} f'(0) = 4a$$

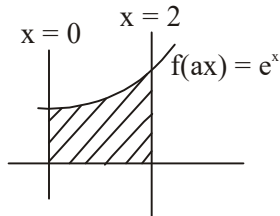
$$\Rightarrow f'(x) = \lambda e^{\lambda x} \Rightarrow \lambda = 4a$$

$$\text{So, } f(x) = e^{4ax}$$

$$f''(x) - 3af'(x) - f(x) = 0$$

$$\Rightarrow \lambda^2 - 3a\lambda - 1 = 0$$

$$\Rightarrow 16a^2 - 12a^2 - 1 = 0 \Rightarrow 4a^2 = 1 \Rightarrow a = \frac{1}{2}$$



$$F(x) = e^{2x}$$

$$\text{Area} = \int_0^2 e^x dx = e^2 - 1$$

155. The area of the region enclosed by the curves $y = x^2 - 4x + 4$ and $y^2 = 16 - 8x$ is :

वक्रों $y = x^2 - 4x + 4$ और $y^2 = 16 - 8x$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है :

- (A) $\frac{4}{3}$ (B) 8 (C) 5 (D) $\frac{8}{3}$

Ans. D

Sol. $y = x^2 - 4x + 4$ $y^2 = 16 - 8x$

$$(x-2)^2 = y$$

$$y^2 = -8(x-2)$$

$$X^2 = Y$$

$$Y^2 = -8X$$

$$\text{Now Area} = \frac{(4a)(4b)}{3} = \frac{8 \times 1}{3} = \frac{8}{3}$$

156. If the area of the region $\{(x, y) : -1 \leq x \leq 1, 0 \leq y \leq a + e^{|x|} - e^{-x}, a > 0\}$ is $\frac{e^2 + 8e + 1}{e}$, then the value of a is:

यदि क्षेत्र $\{(x, y) : -1 \leq x \leq 1, 0 \leq y \leq a + e^{|x|} - e^{-x}, a > 0\}$ का क्षेत्रफल $\frac{e^2 + 8e + 1}{e}$ है, तो a का मान है :

(A) 7

(B) 8

(C) 5

(D) 6

Ans. C

Sol. $\{(x, y) : -1 \leq x \leq 1, 0 \leq y \leq a + e^{|x|} - e^{-x}, a > 0\}$

$$\text{Now, } y \leq a + e^{|x|} - e^{-x}$$

$$\text{C-I : } x \geq 0$$

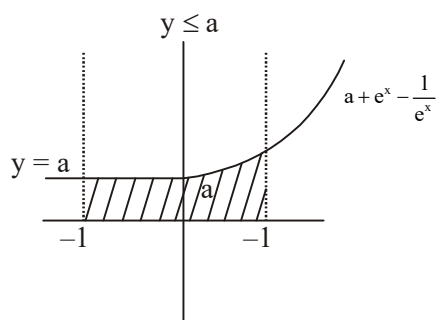
$$y \leq a + e^x - e^{-x}$$

$$y \leq a + e^x - \frac{1}{e^x}$$

$$\text{C-II : } x < 0$$

$$y \leq a + e^{-x} - e^x$$

$$y \leq a$$



∴ A.T.Q

$$\int_{-1}^0 a dx + \int_0^1 (a + e^x - e^{-x}) dx = \frac{e^2 + 8e + 1}{e}$$

$$ax \Big|_{-1}^0 + ax + e^x + e^{-x} \Big|_0^1$$

$$a[0 - (-1)] + (a + e + e^{-1}) - (2) = \frac{e^2 + 8e + 1}{e}$$

$$2a + e + \frac{1}{e} - 2 = \frac{e^2 + 8e + 1}{e}$$

$$2a = \frac{e^2 + 8e + 1}{e} - \left(\frac{e^2 - 2e + 1}{e} \right)$$

$$= \frac{e^2 + 8e + 1 - e^2 + 2e - 1}{e}$$

$$a = 5$$

157. Consider the region $R = \left\{ (x, y) : x \leq y \leq 9 - \frac{11}{3}x^2, x \geq 0 \right\}$. The area, of the largest rectangle of sides parallel to the coordinate axes and inscribed in R , is :

क्षेत्र $R = \left\{ (x, y) : x \leq y \leq 9 - \frac{11}{3}x^2, x \geq 0 \right\}$ पर विचार करें। निर्देशांक अक्षों के समान्तर भुजाओं वाले और R के अंतर्गत अंकित सबसे बड़े आयत का क्षेत्रफल है :

(A) $\frac{821}{123}$

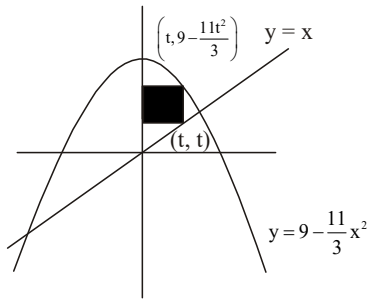
(B) $\frac{730}{119}$

(C) $\frac{625}{111}$

(D) $\frac{567}{121}$

Ans. D

Sol.



$$A = t \left(9 - \frac{11t^2}{3} - t \right)$$

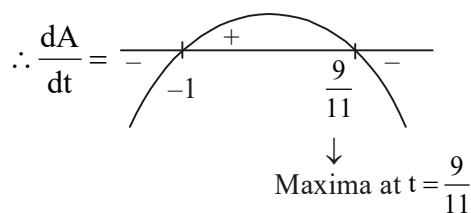
$$A = 9t - t^2 - \frac{11}{3}t^3$$

$$\frac{dA}{dt} = 9 - 2t - 11t^2$$

$$\Rightarrow 11t^2 + 2t - 9 = 0$$

$$11t^2 + 11t - 9t - 9 = 0$$

$$t = -1 \text{ \& } t = \frac{9}{11}$$



$$\therefore \text{largest area} = \frac{9}{11} \left(9 - \frac{11}{3}, \frac{81}{121} - \frac{9}{11} \right)$$

$$= \frac{9}{11} \cdot \frac{63}{11} = \frac{567}{121}$$

158. The area of region $\{(x, y) : x^2 + 4x + 2 \leq y \leq |x + 2|\}$ is equal to :

क्षेत्र $\{(x, y) : x^2 + 4x + 2 \leq y \leq |x + 2|\}$ का क्षेत्रफल बराबर है :

- (A) 5 (B) 20/3 (C) 7 (D) 24/5

Ans. B

Sol. $x^2 + 4x + 2 \leq y \leq |x + 2|$

The area bounded between

$$y = x^2 + 4x + 2 = (x + 2)^2 - 2$$

and $y = |x + 2|$ is same as area bounded between $y = x^2 - 2$ and $y = |x|$

For Point of Intersection $|x|^2 - 2 = |x|$

$$\Rightarrow |x| = 2 \Rightarrow x = \pm 2$$

$$\therefore \text{Required area} = -\int_{-2}^2 (x^2 - 2) dx + \int_{-2}^2 |x| dx$$

$$= -2 \int_0^2 (x^2 - 2) dx + 2 \int_0^2 x dx$$

$$= -2 \left[\frac{x^3}{3} - 2x \right]_0^2 + 2 \left[\frac{x^2}{2} \right]_0^2$$

$$= -2 \left[\frac{8}{3} - 4 \right] + 2 \left[\frac{4}{2} \right]$$

$$= -2 \times \left(\frac{-4}{3} \right) + 4$$

$$= \frac{20}{3}$$

159. The area of the region enclosed by the curves $y = e^x$, $y = |e^x - 1|$ and y -axis is :

चक्रों $y = e^x$, $y = |e^x - 1|$ और y -अक्ष से घिरे क्षेत्र का क्षेत्रफल है :

- (A) $\log_e 2$ (B) $1 + \log_e 2$ (C) $1 - \log_e 2$ (D) $2 \log_e 2 - 1$

Ans. C

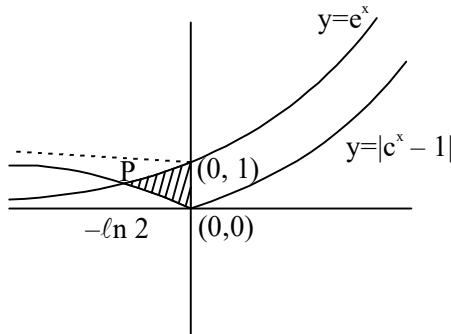
Sol. P : $e^x = -(e^x - 1)$

$$e^x = 1/2$$

$$x = -\ln 2$$

$$\text{Area} = \int_{-\ln 2}^0 e^x - (-e^x + 1)$$

$$= (2e^x - x)_{-\ln 2}^0 = 1 - \ln 2$$



160. The area (in sq. units) of the region $\{(x, y) : 0 \leq y \leq 2|x| + 1, 0 \leq y \leq x^2 + 1, |x| \leq 3\}$ is :

क्षेत्र $\{(x, y) : 0 \leq y \leq 2|x| + 1, 0 \leq y \leq x^2 + 1, |x| \leq 3\}$ का क्षेत्रफल (वर्ग इकाई में) है :

- (A) $\frac{64}{3}$ (B) $\frac{17}{3}$ (C) $\frac{32}{3}$ (D) $\frac{80}{3}$

Ans. A

Sol. Required area = $2 \left[\int_0^2 (x^2 + 1) dx + \int_2^3 (2x + 1) dx \right]$

$$= \frac{64}{3}$$